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The Genetic Basis for Bacterial Mercury Methylation

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Methylmercury is a potent neurotoxin produced in natural environments from inorganic mercury by anaerobic bacteria. However, until now the genes and proteins involved have remained unidentified. Here, we report a two-gene cluster, *hgcA* and *hgcB*, required for mercury methylation by *Desulfovibrio desulfuricans* ND132 and *Geobacter sulfurreducens* PCA. In either bacterium, deletion of *hgcA*, *hgcB* or both genes abolishes mercury methylation. The genes encode a putative corrinoid protein, HgcA, and a 2[4Fe-4S] ferredoxin, HgcB, consistent with roles as a methyl carrier and an electron donor required for corrinoid cofactor reduction, respectively. Among bacteria and archaea with sequenced genomes, gene orthologs are present in confirmed methylators but absent in non-methylators, suggesting a common mercury methylation pathway in all methylating bacteria and archaea sequenced to date.

Mercury (Hg) is a pervasive global pollutant which, as methylmercury (CH_3Hg^+), bioaccumulates in the food web and is highly toxic to humans and other organisms (*1*). Unlike inorganic forms of Hg, which originate from atmospheric deposition and point discharges, methylmercury is generated in the environment predominantly by anaerobic microorganisms (*2*). Sulfate-reducing bacteria are the main producers of CH_3Hg^+ (*3*, *4*), although iron-reducing bacteria (*5*–*7*) and methanogens (*8*, *9*) can also be involved.

Production of CH_3Hg^+ by the model methylating bacteria *Desulfovibrio desulfuricans* ND132 and *Geobacter sulfurreducens* PCA involves cellular uptake of Hg(II) by active transport, methylation of Hg(II) in the cytosol, and export of CH_3Hg^+ from the cell (*10*). Hg methylation is an enzyme-catalyzed process proposed to be associated with the reductive acetyl-CoA (Wood-Ljungdahl) pathway (*11*) and potentially linked to corrinoid proteins involved in this pathway (*12*). However, no direct evidence firmly connects the acetyl-CoA pathway and the ability of bacteria to methylate Hg (*13*). Furthermore, phylogenetic analyses have not revealed any distinctive trends or clustering of methylating versus non-methylating microorganisms (*14*–*16*).

To understand the genetic and biochemical basis of microbial Hg methylation, we analyzed the genomes of methylating and non-methylating bacteria in the context of biochemical pathways involved in single carbon metabolism. The well characterized corrinoid iron-sulfur protein (CFeSP) is known to transfer methyl groups to a NiFeS cluster in acetyl-CoA synthase (*17*). Therefore, recognizing that a corrinoid protein associated with the acetyl-CoA pathway could be required for Hg methylation, we reasoned that a protein similar to CFeSP might transfer a methyl group to a Hg substrate to yield CH_3Hg^+, and that genes encoding such a protein should be recognizable in the genome sequences of Hg methylating bacteria. Complete genome sequences are available for six methylating and eight closely related non-methylating bacterial species (tables S1 and S2). Furthermore, molecular structures and functions have been determined for various enzymes of the reductive acetyl-CoA pathway, including CFeSP from *Moorella thermoacetica* (*18*, *19*) and *Carboxydothermus hydrogenoformans* (*20*, *21*). Accordingly, we performed a BLASTP search with the sequence of the large subunit of CFeSP (CfsA, locus tag CHY_1223) from *C. hydrogenoformans* Z-2901 against the translated genome sequence of *D. desulfuricans* ND132 (*22*). Sequence similarity was found between the C-terminal corrinoid-binding domain of CfsA and the N terminus of DND132_1056 (fig. S1), although DND132_1056 lacks both the TIM barrel domain and the C-terminal [4Fe-4S] binding motif of CfsA. The C-terminal region showed no detectable similarity to any proteins of known structure, but exhibited features characteristic of a transmembrane domain (fig. S2).

We also performed comparative genomic analyses of known Hg methylators and non-methylators based on Pfam classifications (*23*), with an emphasis on enzyme families known to be involved in methyl transfer reactions. The distribution of Pfam domains in the genomes is heterogeneous and, for the most part, does not coincide with the mercury methylation phenotype (table S3). However, the distribution of proteins of the CdhD family (PF03599, annotated as CO dehydrogenase/acetyl-CoA synthase delta subunit) encoded in the genomes correlates with the ability or inability of those organisms to methylate mercury. DND132_1056 is annotated as encoding a CdhD member, as are its close relatives in all five other confirmed methylators.

Analysis of the genomic context in the confirmed Hg methylators revealed genes similar to both the putative corrinoid protein-encoding gene and an additional, ferredoxin-like gene located downstream, suggesting that these two genes might be co-expressed and functionally related (Fig. 1). In *D. desulfuricans* ND132, the annotated coding sequences of the two genes (DND132_1056 and DND132_1057) are on the same strand and are separated by only 14 bp. Similar gene pairs were found in the genomes of 52 organisms with sequence translations available in public databases (table S4). The two genes are present in all sequenced, confirmed methylators and absent in the sequenced, confirmed non-methylators. The other 46 organisms in which the genes are present have not been tested for Hg methylation (table S4).

We hypothesized that these two genes are key components of the bacterial Hg methylation pathway, with the putative corrinoid protein facilitating methyl transfer and the ferredoxin carrying out corrinoid reduction. Therefore, we deleted these genes individually, and also together, from *D. desulfuricans* ND132 (supplementary text). Additionally, we deleted the orthologs GSU1440 and GSU1441 together, and GSU1440 individually, from *G. sulfurreducens* PCA. In both of these organisms, CH_3Hg^+ production decreased in the deletion mutants by greater than 99% relative to the parental strains (Fig. 2). Complementation of the two-gene deletions by reincorporation of the genes into the

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chromosomes restored 26% and 87% of the wild-type methylation activity in *D. desulfuricans* ND132 and *G. sulfurreducens* PCA, respectively, as measured by ICP-MS (Fig. 2). Deletion of DND132_1057 alone yielded <0.2% of wild-type methylation activity and subsequent complementation showed 97% methylation activity. Complementation of either gene alone into the double deletion mutant did not restore detectable methylation activity (Fig. 2). Restoration of ΔDND132_1056 was not performed. Although the relative location of the two genes is consistent with co-transcription, RT-PCR confirmed transcription of DND132_1056 in the ΔDND132_1057 strain and DND132_1057 in the ΔDND132_1056 mutant (fig. S3).

The above findings are consistent with both genes being required for Hg methylation activity, although other unidentified genes are also likely to be involved. Hereafter, we refer to the DND132_1056 gene and its inferred orthologs as *hgcA*, encoding putative corrinoid proteins required for CH_3Hg^+ production, and DND132_1057 and its inferred orthologs as *hgcB*, encoding putative corrinoid protein-associated 2[4Fe-4S] ferredoxins. To determine whether gene loss impairs metabolism on a more general scale, we obtained comparative growth curves. The deletion mutants showed no impairment in rate or extent of growth (fig. S4). Thus, under the conditions tested, the construction of the deletions did not cause major growth aberrations that might interfere with the detection of methylation activity. The native functions of *hgcA* and *hgcB* remain unknown, but these genes are not essential for cell survival or proliferation.

The above findings merit some mechanistic considerations. The requirement for HgcA and HgcB in methylation is largely consistent with a previously proposed Hg methylation pathway by Bartha and coworkers (*11*), which we revise (fig. S5 and supplementary text). The methyl group in CH_3Hg^+ originates from CH_3-H_4folate in *D. desulfuricans* LS (*11*) and is likely first transferred (as CH_3^+) to cob(I)alamin-HgcA to form CH_3-cob(III)alamin-HgcA. This step may be catalyzed by a folate-binding methyltransferase similar to the CH_3-H_4folate:CFeSP methyltransferase (MeTr) from the reductive acetyl-CoA pathway (*24*, *25*), or by an unknown enzyme (fig. S5 and supplementary text). Based on the high affinity of thiolate ligands for Hg^{2+} (formation constants for $Hg(SR)_2$, log K = 40-43) (*26*), a possible substrate for HgcA could be a Hg(II) *bis* thiolate complex involving either free cellular thiols or cysteine residues from a protein. Methyl transfer from CH_3-cob(III)alamin-HgcA to a Hg substrate likely involves either $CH_3\cdot$ or CH_3^-. Although transfer of a carbanion from methylcobalamin to Hg(II) non-enzymatically is known to occur (*27*, *28*), enzymatic transfer of CH_3^- by a corrinoid protein has never been observed.

Further sequence analysis of the 52 HgcA orthologs (Fig. 3) revealed a highly conserved motif, N(V/I)WCA(A/G)GK, in the region of highest similarity to the CfsA subunit of CFeSP. This region corresponds to the cap helix of CFeSP, which is located near the lower-axial face of the corrin ring (*20*). In all HgcA sequences, a strictly conserved cysteine (Cys93 in *D. desulfuricans* ND132) occupies the position corresponding to Thr374 in CFeSP from *C. hydrogenoformans* Z-2901 (Fig. 3 and fig. S1). Although Thr374 is not considered a ligand for Co (*19*, *20*), a cysteine might coordinate to Co, depending on its location relative to the cofactor and the Co oxidation state. Homology modeling (fig. S6) and UV/Vis spectra (fig. S7) of the cobalamin-binding domain of HgcA suggest that Co-S coordination may be present in HgcA. Lower-axial coordination of alkyl-cob(III)alamin by a biological thiolate has been proposed previously (*29*), but never observed for a corrinoid protein. The likely role of the ferredoxin-like protein HgcB is to accomplish the thermodynamically difficult reduction of Co(II) to Co(I) required for turnover, consistent with a previously suggested need for a ferredoxin as a reductant in Hg methylation (*11*). The mechanistic details (supplementary text) of methyl transfer, the integration of these two gene products into carbon and energy metabolism, and their functioning with other potential, yet unidentified, proteins remain to be determined.

In the absence of genome sequences for all Hg methylating organisms, the generality of the present findings cannot yet be ascertained. However, our interpretation is in agreement with all currently available sequence information for methylating bacteria and archaea. The presence of the *hgcAB* cluster in the genomes of several sequenced, but so far untested, microorganisms (table S4) leads us to hypothesize that these organisms are also capable of methylating mercury. The gene cluster appears to be quite sporadically distributed across two phyla of bacteria (*Proteobacteria* and *Firmicutes*) and one phylum of archaea (*Euryarchaeota*). Organisms possessing the two-gene cluster include 24 strains of *Deltaproteobacteria*, 16 *Clostridia*, one *Negativicutes*, and 11 *Methanomicrobia*. Interestingly, we also found these genes in a psychrophile (*30*), thermophile (*31*), and in a human commensal methanogen (*32*) (Fig. 3). The sparse phylogenetic gene distribution of the *hgcAB* system may be due to gene loss, lateral gene transfer, or both, across distant taxa and may be linked to environmental and community structure factors. The sporadic distribution of these genes and the lack of an obvious selective advantage related to mercury toxicity (*15*) raise important questions regarding their physiological roles. Identification of these genes is a critical step linking specific microorganisms and environmental factors that influence microbial Hg methylation in aquatic ecosystems.

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Supplementary Materials
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Materials and Methods
Supplementary Text
Figs. S1 to S9
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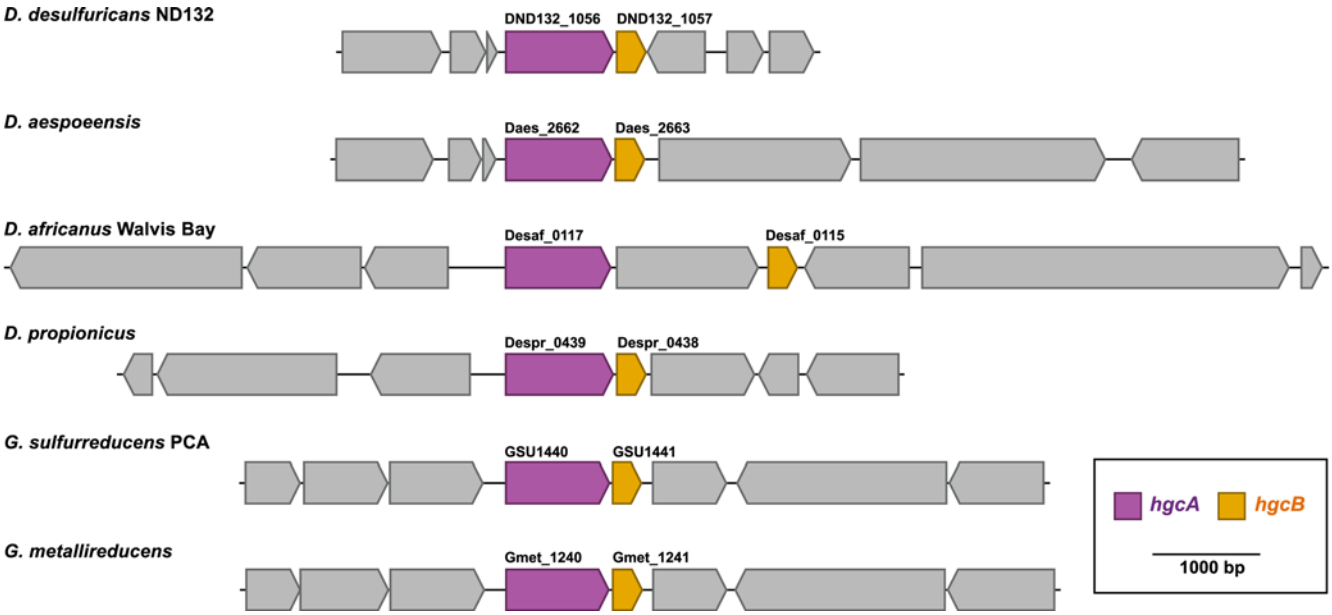


Fig. 1. Putative mercury methylation gene cluster and genomic context for six confirmed mercury methylators with sequenced genomes.

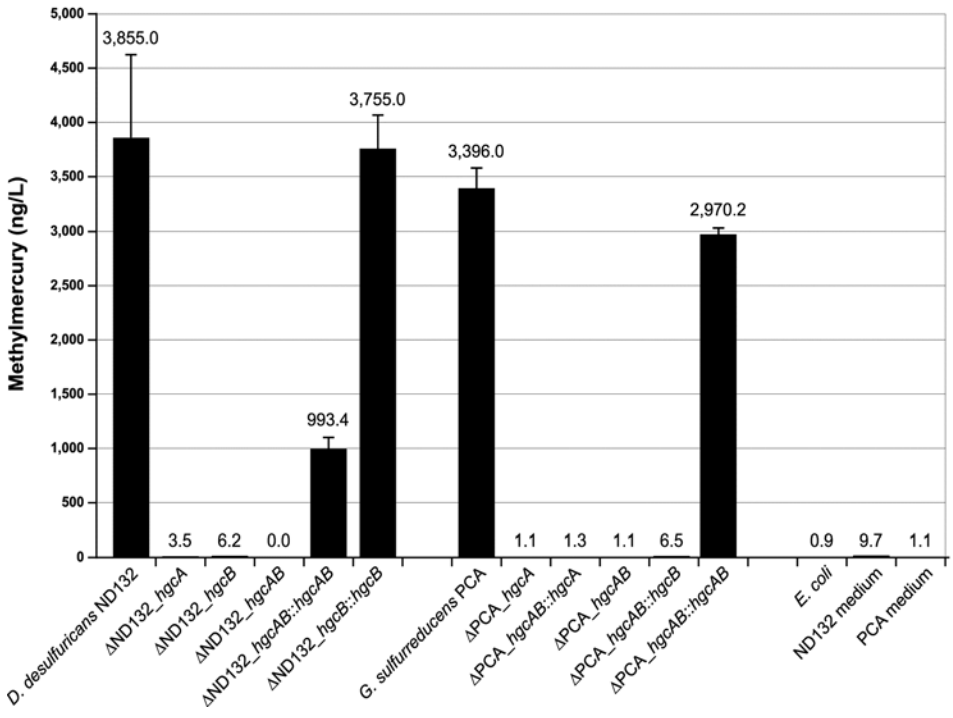


Fig. 2. Production of methylmercury by *D. desulfuricans* ND132 and *G. sulfurreducens* PCA, by deletions of *hgcA, hgcB* and *hgcAB*, and by complements in both bacteria. CH_3Hg^+ concentrations (ng/L) were determined after overnight incubations and measured by ICP-MS. A "Δ" prefix indicates a gene deletion and "**::**" indicates complementation by chromosomal insertion. Values plotted are the average CH_3Hg^+ concentrations detected per strain from triplicate assays, and error bars are standard deviations. It should be noted that ΔPCA_*hgcAB* complemented with PCA_*hgcA*$^+$ or PCA_*hgcB*$^+$ is still deleted for PCA_*hgcB* or PCA_*hgcA*, respectively.

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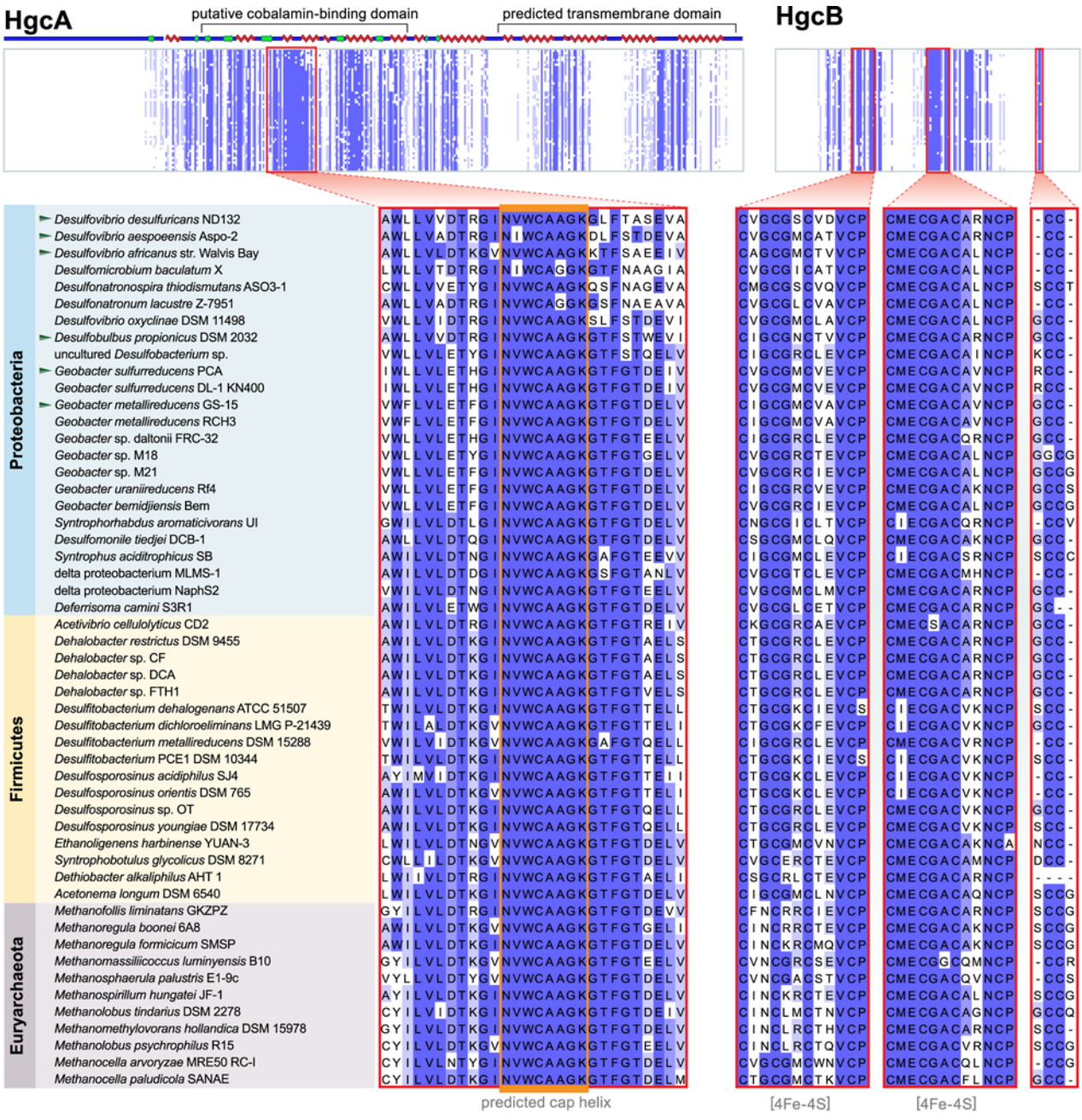


Fig. 3. Multiple sequence alignments of 52 HgcA and HgcB orthologs from bacteria and archaea including six confirmed methylating bacteria (indicated by arrows). Red boxes indicate highly conserved regions including the putative cap helix (consensus sequence motif, N(V/I)WCA(A/G)GK) in HgcA, two strictly conserved $CX_2CX_2CX_3C$ motifs characteristic of [4Fe-4S] clusters, and a conserved vicinal pair of cysteines located at the C terminus of HgcB.

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